

# Group-Fair Stochastic Gradient Descent via Masked Optimal Transport

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Table 1: stub

**Lemma 1.** *stub*

**Lemma 2.** *stub*

**Theorem 3.** *stub*

**Theorem 4.** *stub*

**Corollary 5.** *stub*

**Theorem 6.** *stub*

(1)

(2)

**Proposition 7.** *stub*

## A APPENDIX

**A. Proofs.** *Theorem 6 (sketch).* Route supply  $p_i$  from a source through group  $i$ , along the edges of  $E$  with infinite capacity, into composition  $j$  with demand  $q_j$ , and to a sink. A coupling exists iff the maximum flow saturates every source edge, which by max-flow/min-cut [?] is the right inequality of (1); the left one is the same statement for the reversed network. On singletons (1) is Hall’s condition [?].

*Corollary 5.* For  $I = \{i\}$ ,  $\mathcal{N}(I)$  is the set of compositions containing  $i$ , of total mass  $r_i$ , and  $\mathcal{S}(I) = \{j : A_j = \{i\}\}$ . Substitute  $r_i = 1 - (1 - \pi_i)^b$  and  $r_i = K/N$ .

*Theorem 3.* By non-expansiveness of  $\Pi_C$ ,  $\|\theta^t - \theta^*\|^2 \leq \|\theta^{t-1} - \theta^*\|^2 - 2\eta\langle \hat{g}^t, \theta^{t-1} - \theta^* \rangle + \eta^2\|\hat{g}^t\|^2$ . Conditioning on  $\theta^{t-1}$ , Lemma 1 and convexity bound the inner product below by  $F_p(\theta^{t-1}) - F_p(\theta^*)$ , and Lemma 2 bounds the last term by  $\eta^2 G^2$ . Summing, telescoping, and applying Jensen to  $\bar{\theta}_T$  gives  $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$ , and the stated  $\eta$  balances the two terms [?]. In full:  $\mathbb{E}\|\theta^t - \theta^*\|^2 \leq \mathbb{E}\|\theta^{t-1} - \theta^*\|^2 - 2\eta\mathbb{E}[F_p(\theta^{t-1}) - F_p(\theta^*)] + \eta^2 G^2$ ; summing over  $t \leq T$ , telescoping, dropping the terminal distance and applying Jensen to  $\bar{\theta}_T$  gives  $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$ , which is  $DG/\sqrt{T}$  at  $\eta = D/(G\sqrt{T})$ ; the same bound holds for  $\eta_t = \eta_0/\sqrt{t}$  up to constants.

*Theorem 4.* Add and subtract  $F_p(\theta^\dagger) = \Phi(\hat{p})$ ; the floor is exact by definition and  $\Phi(p) = F_p(\theta^*)$ . Theorem 3 applied to  $F_{\hat{p}}$  gives the residual bound on the surrogate; the residual on  $F_p$  differs from it by  $(F_p - F_{\hat{p}})(\bar{\theta}_T) - (F_p - F_{\hat{p}})(\theta^\dagger)$ , which vanishes as  $\bar{\theta}_T \rightarrow \theta^\dagger$  (for instance under strong convexity, which both of our instances satisfy). The  $\ell_1$  bound is the two-point argument of Proposition 7 with  $w$  for  $\hat{p}$ .

**B. The infeasible regime in closed form (IMDb-Wiki,  $\beta = 3$ ).** On the mirrored instance, where 41 of 100 groups violate  $p_i \leq r_i$  and hold 88% of the importance mass, the Sinkhorn limit truncates the target at the observation-rate ceiling,  $\hat{p}_i \approx \min(p_i, r_i)$  up to renormalization (maximum deviation 0.023), giving  $\|p - \hat{p}\|_1 = 1.58$  of a maximum possible 2. Because the task is linear regression, the minimizers  $\theta^*$  of  $F_p$  and  $\theta^\dagger$  of  $F_{\hat{p}}$  are available in closed form:  $\Phi(p) = F_p(\theta^*) = 82.9$  (matching the measured full-coverage plateau, 82.95) and  $\Phi(\hat{p}) = F_{\hat{p}}(\theta^\dagger) = 105.9$ . The alignment floor is therefore 23.0, and the realized bias equals its exact expression  $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$ , where  $L(\theta)$  collects the per-group risks. The  $\ell_1$  form  $2M\|p - \hat{p}\|_1$  of Theorem 4 holds but is conservative here, since any uniform bound satisfies  $M \geq \max_i L_i(\theta^\dagger) \approx 454$ . The group-blind surrogate  $\tilde{p}_i = r_i/K$  has the same floor to two decimals ( $\Phi(\tilde{p}) = 105.9$ ): both surrogates starve the same groups, which is why  $\Delta_{\text{floor}}$  vanishes at this endpoint of Table 1.

**C. Adult severity sweep.** The five Adult skews omitted from Table 1, same conventions.

| $\beta$ | $\nu$ | $\Delta_{\text{floor}}$ | gain   | $t$  |
|---------|-------|-------------------------|--------|------|
| 0       | .60   | .0110                   | +.0210 | 44.7 |
| 0.25    | .60   | .0144                   | +.0181 | 69.4 |
| 0.5     | .40   | .0129                   | +.0134 | 35.2 |
| 0.75    | 0     | .0049                   | +.0050 | 35.8 |
| 1       | 0     | .0003                   | +.0002 | 7.1  |

**D. Adult per-race cross-entropy, prevalence regime.** Tail-500 over 5 seeds, decaying stepsize.

|                  | White       | Black       | Asian-Pac.  | Amer-Ind.   | Other       |
|------------------|-------------|-------------|-------------|-------------|-------------|
| GAVOT            | .339        | <b>.179</b> | <b>.368</b> | <b>.175</b> | <b>.073</b> |
| group-blind avg. | <b>.332</b> | .200        | .399        | .189        | .119        |
| full             | .379        | .177        | .328        | .123        | .031        |

Transport improves four of five categories and regresses only the majority group, the expected effect of reweighting toward a category-uniform target; the residual loss on the two smallest categories is the unreachable mass of Cor. 5.